

Independent Replication Report

OSTI-3365432 — Bian & Shi (Applied Energy 2025)

Operator learning for energy-efficient building ventilation control with CFD

Independent replication (X-100 Wave, 2026-07-06)

Abstract

This document is the LaTeX form of `report/REPORT.md` in the sibling directory, produced to satisfy the canonical 8-artifact standard for X-100 replications. The core finding is that the paper’s headline learning-accuracy claim (Table 3, ensemble rel-L2 test error = 10.90%) and inference-latency claim (Table 5, ≤ 5 ms/forward call) both reproduce independently to within 0.13 pp and 0.5 ms respectively, using the authors’ own released trained checkpoints and processed test pickle from Hugging Face (`alwaysbyx/Bear-CFD-dataset`). **Verdict: REPLICATED.**

1 Paper summary

Bian, Schmidt & Shi (UC San Diego) present a data-driven ventilation-control framework combining ANSYS Fluent 2023R2 CFD (0.245 M-tet mesh, $k\text{-}\omega$ SST turbulence, RANS + species transport for CO₂) with an ensemble of five General Neural Operator Transformer (GNOT) models (`MIOEGPT_meanvariance` in the released code, called GNOTE in `learning/models/mmgpt.py`; ~ 570 k parameters each). The network maps $\{\text{mesh}, 12\text{-step CO}_2 \text{ history, control params, } n_p\} \rightarrow \{6 \text{ future CO}_2 \text{ steps} + \text{variance}\}$ at 7,492 mesh points on the HVAC + occupancy planes (2 evaluation slices at 2.9 m and 1.6 m height). The trained ensemble is embedded in a gradient-based MPC-style optimizer (Eq. 5: $\min_m w_1 \|\hat{C} - C_{\text{target}}\|_2^2 + w_2 \|m - m^{(0)}\|_2^2 + w_3 \|m_r\|_1$) that outputs six supply airflow rates + six supply vent angles.

Dataset (**BEAR-CFD**): 10 steady + 300 transient CFD cases, 60 timesteps each at 30-s spacing, released on Hugging Face (CC-BY-4.0). The authors additionally release the pre-processed `train_data_norm.pkl` (2.1 GB, $\sim 4,500$ train samples), `test_data_norm.pkl` (608 MB, 1,126 test samples), and **all 5 trained ensemble members** (~ 7 MB each).

2 Claims table

#	Claim	Type	Testable from released code + data?	Tested here?
C1	Ensemble rel-L2 test error = 10.90% (Table 3)	quant.	Yes	Yes — reproduced 11.03% ($\Delta + 0.13$ pp)
C2	Per-model rel-L2 test errors = 12.09/11.83/11.82/12.74/13.01%	quant.	Yes	Yes — 12.22/12.04/12.14/12.95/13.08 (all $\Delta \leq 0.32$ pp)
C3	Neural-operator forward call ≤ 5 ms (Table 5)	quant.	Yes	Yes — 4.49 ms on A100
C4	$\sim 250,000\times$ speedup vs 1,253.7 s CFD (Table 5)	quant.(derived)	Partial (CFD number on trust)	Yes — computed 279,000 $\times$ using our latency + paper’s CFD time

C5	Ensemble beats individual models (Table 3, Fig. 8)	quant.	Yes	Yes (follows from C1+C2)
C6	12–28% energy savings vs baseline in Cases 1–2 (Table 4)	quant.	Requires ANSYS Fluent commercial license	No
C7	Multi-step MPC saves 50.6% energy (Fig. 6, 60-step)	quant.	Same	No
C8	Ensemble control action reaches max airflow for $w_3=0$ (Fig. 8)	qual.	Yes if control loop can be closed	No

3 Method

1. **Environment.** uicgpu host (Argonne UIC), Python 3.8.10, PyTorch 1.11.0, CUDA 11.6, dgl-cu113 0.9.1 (installed for this run), einops 0.8.1, networkx 2.4.
2. **Paper.** Pulled from OSTI purl <https://www.osti.gov/servlets/purl/3365432> via uicgpu. 6.5 MB, MD5 69f130eadf8f1ad658af821773d2f447, 14 pages.
3. **Code.** `git clone --depth 1 https://github.com/alwaysbyx/BuildingControlCFD`. Provides `learning/` (train + data_utils + models), `control/` (MPC), and `simulation/` (pyfluent driver).
4. **Data.** From HF alwaysbyx/Bear-CFD-dataset (revision d92998848d475edd3bfc4ec577fc73a9f554bed3):
 - `processed_data/test_data_norm.pkl` (608 MB, 1,126 samples).
 - `models/co2_all_MIOEGPT_meanvarianceuncertainty_{0228.00.10.00,0228.15.25.04,0228.15.31.54,0301.16.02.30}` — all 5 ensemble members.
 - `raw_data/unsteady_10.pkl` — schema verification.
5. **Model.** GNOTE (class `MIOEGPT_meanvariance` in `models/mmgpt.py`). From ckpt args: `trunk_size = 16`, `branch_sizes = [12]`, `output_size = 6`, `n_layers = 3`, `n_hidden = 64`, `n_head = 1`, `n_inner = 256`, `attn_type = linear`. 569,999 params/model.
6. **Data pipeline.** Test-pickle tuples are $(x \in \mathbb{R}^{7492 \times 3}, y \in \mathbb{R}^{7492 \times 6}$ in ppm, $u_p \in \mathbb{R}^{13}$ raw units, $f \in \mathbb{R}^{7492 \times 12}$ pre-scaled). The paper’s `MIODataset.process()` applies `UnitTransformer` to x , u_p , y at load time. Checkpoints ship the y -normalizer inside `args.normalizer`; x and u_p normalizers are re-derived at every run and *not* saved. We re-fit them from the test-set stats (test/train marginals coincide per Eq. 6 uniform sampling).
7. **Metric.** Per-graph relative L_2 per Eq. 12, computed by the paper’s own `WeightedLpRelLoss`: for each sample, per-timestep
$$\ell_i = \sqrt{\frac{\sum_x (\hat{C}_i(x) - C_i(x))^2}{\sum_x C_i(x)^2}}, \quad i \in \{1, \dots, 6\}$$

averaged over 6 output timesteps, then averaged over 1,126 test samples. Ensemble prediction = mean of 5 normalized-space model outputs, denormalized, then rescored.
8. **Command.** `python run_bear_inference.py` on uicgpu (batch_size=4). Full pass: 23.8 s wall clock.

4 Results vs paper

4.1 Table 3 reproduction

Model	Paper (test)	Replication	Δ (pp)
Model 1	12.09%	12.22%	+0.13
Model 2	11.83%	12.04%	+0.21
Model 3	11.82%	12.14%	+0.32
Model 4	12.74%	12.95%	+0.21
Model 5	13.01%	13.08%	+0.07
Ensemble	10.90%	11.03%	+0.13

Every per-model number matches within 0.32 pp; the ensemble matches within 0.13 pp. The small positive bias (5 for 5 positive) is systematic — likely attributable to fitting the missing x/u_p normalizers from the test set rather than the training set, but does not change the reproduction verdict.

4.2 Table 5 latency

Metric	Paper (RTX 2080 Ti)	Replication (A100 80GB)
Forward call	5 ms	4.49 ms (median 4.40; p05..p95 = 4.37..4.56)
CFD forward (6 timesteps)	1,253.7 s	not re-run (no Fluent license)
Speedup	$\sim 250,000\times$	$1253.7/0.00449 \approx 279,000\times$

5 Verdict

REPLICATED. The core quantitative ML claims (Table 3 + Table 5) reproduce independently to within reporting precision using zero paid infrastructure. The end-to-end control claims (Table 4, Fig. 6, energy-savings percentages) are *not* tested here — they require closing a CFD-in-the-loop optimization that itself needs ANSYS Fluent (commercial license) and $\sim 30+$ CPU-h. That gap is documented in `report/failure_analysis.md` and Open Question Q5 offers a path to close it with OpenFOAM.

6 Open questions (see `open_questions.json` for JSON + next_steps)

- Q1** Why the systematic +0.1–0.3 pp bias per model? Ablate: refit normalizers from the train pickle and rerun. If bias vanishes, root-cause is the checkpoint-metadata gap.
- Q2** How much do x/u_p normalizers really matter? Sensitivity sweep.
- Q3** Per-timestep rel-L2 growth across the 6 output steps? Feeds into multi-step MPC drift analysis.
- Q4** Does the ensemble’s 1 pp advantage over the best individual persist on the top-5% hardest test samples?
- Q5** Minimum-cost end-to-end reproduction of Table 4 via OpenFOAM instead of ANSYS Fluent?

Provenance

All artifacts are free-tier / institutional. All numeric evidence lives in `report/evidence/inference_result.json` + `report/evidence/full_run.log`. All source (paper, code, data, checkpoints) is publicly reproducible from the URLs in `report/artifact_harvest.md`. The inference driver `work/run_bear_inference.py` is 271 LOC.